

Example: Finding the Optimal Quantity of Leisure, Consumption, and Labor Supply to Maximize Utility Subject to Budget and Time Constraints*

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December 28, 2023

I The Problem

Suppose the representative consumer is given a choice between consumption (c), leisure (l), and labor supply (N^s). The representative consumer's utility function can be written as $U(c, l) = c^{\frac{2}{3}}l^{\frac{1}{3}}$. The representative consumer's real wage (w) is equal to 3 units of consumption goods, dividend income (Π) is equal to 4 units of consumption goods, and taxes (T) equal to 10 consumption goods. The representative consumer has 24 hours to allocate between leisure and labor (h). We want to:

- Find the optimal quantity of consumption and leisure (c, l) to
- Maximize utility $U(c, l) = c^{\frac{2}{3}}l^{\frac{1}{3}}$ subject to the budget constraint and
- Find the optimal level of labor supply N^s subject to the time constraint

*This is a problem from ECON 32041: Intermediate Macroeconomic Theory and Policy instructed by Dr. Duhyeong Kim

II Setup

There are 3 conditions that must be met to find the optimal levels:

1. $MRS = w$
2. The budget constraint
3. The time constraint

Where MRS is the Marginal Rate of Substitution.

First, we will find the budget constraint. The budget constraint states that:

$$c = -wl + wh + \Pi - T$$

Plugging in real wage (w), dividend income (Π), taxes (T), and hours (h):

$$c = -3l + 3(24) + 4 - 10 \Rightarrow c = -3l + 66$$

The time constraint states that:

$$l + N^s = h$$

Plugging in hours (h):

$$l + N^s = 24$$

We will use the Lagrangian Function to find the optimal combination of consumption and leisure. The Lagrangian Function can be defined as:

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

Here we use the Lagrange Multiplier (λ). Our application of the Lagrangian Function states that:

$$\mathcal{L}(c, l, \lambda) = U(c, l) - \lambda(c + 3l - 66)$$

Here we rewrite the function $c = -3l + 66$ as $c + 3l - 66 = 0$ in order to maximize utility.

Expanding our utility function:

$$\mathcal{L}(c, l, \lambda) = c^{\frac{2}{3}}l^{\frac{1}{3}} - \lambda(c + 3l - 66)$$

Next, we distribute the Lagrange Multiplier (λ):

$$\mathcal{L}(c, l, \lambda) = c^{\frac{2}{3}}l^{\frac{1}{3}} - \lambda c + \lambda 3l - \lambda 66$$

Now that we have set up our Lagrangian and constraints, we can move on to the solution.

III Solution

First, we will take the partial derivative of our Lagrangian with respect to consumption (c):

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c} &= \frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} + \lambda c^0 \\ &= \frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} + \lambda\end{aligned}$$

Since $\lambda 3l$ and $-\lambda 66$ do not contain the variable c , they are constants and can be eliminated using the constant rule. The rest of the derivative is found by the power rule.

Second, we take the partial derivative of our Lagrangian with respect to leisure (l):

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial l} &= c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}} + \lambda 3l^0 \\ &= c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}} + \lambda 3\end{aligned}$$

Since λc and $-\lambda 66$ do not contain the variable l , they are constants and can be eliminated using the constant rule. The rest of the derivative is found by the power rule.

Third, we take the partial derivative of our Lagrangian with respect to the Lagrange Multiplier (λ):

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \lambda} &= \lambda^0 c + \lambda^0 3l - \lambda^0 66 \\ &= c + 3l - 66\end{aligned}$$

Since $c^{\frac{2}{3}}l^{\frac{1}{3}}$ does not contain the variable λ , it is a constant and can be eliminated using the constant rule. The rest of the derivative is found by the power rule.

Now that we have found our partial derivatives, the next step is to relate consumption and leisure. Since both terms contain the Lagrange Multiplier (λ), we can set each equal to the Lagrange Multiplier:

$$\begin{aligned}\lambda &= -\frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} \\ \lambda &= -\frac{c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}}}{3}\end{aligned}$$

so

$$\begin{aligned}-\frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} &= -\frac{c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}}}{3} \\ \Rightarrow \frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} &= \frac{c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}}}{3}\end{aligned}$$

Next, we can multiply both sides of the equation by 3:

$$\frac{6}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} = c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}}$$

Reordering the negative exponents and multiplying the fractions on both sides:

$$\begin{aligned}\frac{6}{3} \frac{l^{\frac{1}{3}}}{c^{\frac{2}{3}}} &= \frac{1}{3} \frac{c^{\frac{2}{3}}}{l^{\frac{2}{3}}} \\ \Rightarrow \frac{6l^{\frac{1}{3}}}{3c^{\frac{1}{3}}} &= \frac{c^{\frac{2}{3}}}{3l^{\frac{2}{3}}}\end{aligned}$$

Next, we can cross multiply and solve for c:

$$\begin{aligned}\frac{6l^{\frac{1}{3}}}{3c^{\frac{1}{3}}} &= \frac{c^{\frac{2}{3}}}{3l^{\frac{2}{3}}} \\ \Rightarrow 3c &= 18l \\ \Rightarrow c &= 6l\end{aligned}$$

Now we set our partial derivative with respect to the Lagrange Multiplier (λ) equal to 0 and plug in c:

$$c + 3l - 66 = 0 \Rightarrow 6l + 3l - 66 = 0 \Rightarrow 9l - 66 = 0 \Rightarrow 9l = 66 \Rightarrow l = \frac{66}{9} = \frac{22}{3}$$

Now that we know the value of leisure (l), we can find the value of consumption (c):

$$c = 6l \Rightarrow c = 6 \left(\frac{22}{3} \right) \Rightarrow c = 44$$

To find the optimal quantity of labor supply (N^s), we solve the time constraint by plugging in the value for leisure (l) and solving for N^s :

$$\begin{aligned}l + N^s &= 24 \\ \Rightarrow \frac{22}{3} + N^s &= 24 \\ \Rightarrow N^s &= \frac{50}{3}\end{aligned}$$

Since we know the optimal values of consumption and leisure, we can solve for the Lagrange Multiplier:

$$\begin{aligned}\lambda &= -\frac{2}{3}c^{-\frac{1}{3}}l^{\frac{1}{3}} \\ \lambda &= -\frac{2}{3}(44)^{-\frac{1}{3}}\left(\frac{22}{3}\right)^{\frac{1}{3}} \approx -0.3669\end{aligned}$$

and

$$\begin{aligned}\lambda &= -\frac{c^{\frac{2}{3}}\frac{1}{3}l^{-\frac{2}{3}}}{3} \\ \lambda &= -\frac{(44)^{\frac{2}{3}}\frac{1}{3}\left(\frac{22}{3}\right)^{-\frac{2}{3}}}{3} \approx -0.3669\end{aligned}$$

Therefore, the optimal combination of consumption and leisure is 44 units of consumption and $\frac{22}{3}$ units of leisure. The optimal quantity of labor supply is $\frac{50}{3}$ units. The Lagrange Multiplier is equal to -0.3669 .

IV Verifying The Solution

We can verify our solution by checking that our optimality conditions are met; $MRS = w$, $c = -3l + 66$, and $l + N^s = 24$. A shortcut to finding the MRS is:

$$\frac{MU_l}{MU_c}$$

Where MU_l is the marginal utility of leisure, which we found by taking the partial derivative of the Lagrangian with respect to leisure. MU_c is the marginal utility of consumption, which we found by taking the partial derivative of the Lagrangian with respect to consumption.

Plugging in our marginal utilities and values for consumption and leisure:

$$\begin{aligned} & \frac{c^{\frac{2}{3}} \frac{1}{3} l^{-\frac{2}{3}}}{\frac{2}{3} c^{-\frac{1}{3}} l^{\frac{1}{3}}} \\ & \Rightarrow \frac{\frac{1}{3} c}{\frac{2}{3} l} \\ & \Rightarrow \frac{c}{2l} \\ & \Rightarrow \frac{44}{2 \left(\frac{22}{3} \right)} = 3 = w \end{aligned}$$

Next, we verify our time constraint:

$$\begin{aligned} l + N^s &= 24 \\ \Rightarrow \frac{22}{3} + \frac{50}{3} &= 24 \\ \Rightarrow 24 &= 24 \end{aligned}$$

Finally, we can plug in our values of consumption and leisure to verify the budget constraint is satisfied:

$$c = -3l + 66 \Rightarrow 44 = -3\left(\frac{22}{3}\right) + 66 \Rightarrow 44 = 44$$

It is also possible to verify the optimal levels of consumption and leisure using the following equations, where α is equal to the power of consumption and β is equal to the power of leisure:

$$\begin{aligned} c &= \frac{\alpha(wh + \Pi - T)}{\alpha + \beta} \\ \Rightarrow c &= \frac{\frac{2}{3}(3(24) + 4 - 10)}{\frac{2}{3} + \frac{1}{3}} \\ &\Rightarrow c = 44 \end{aligned}$$

and

$$\begin{aligned}l &= \frac{\beta(wh + \Pi - T)}{w(\alpha + \beta)} \\ \Rightarrow l &= \frac{\frac{1}{3}(3(24) + 4 - 10)}{3(\frac{2}{3} + \frac{1}{3})} \\ \Rightarrow l &= \frac{22}{3}\end{aligned}$$